Analyst Lab Africa

Experience Lab Internship

HealthConnect Rwanda Healthcare Analytics & Artificial Intelligence Transformation:

Executive Business Intelligence Report

Data Analytics Track

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Healthcare Operations and Performance Analytics Dashboard
Power BI Solution Summary & Strategic Recommendations

Week 6: EXECUTIVE BUSINESS INTELLIGENCE REPORT

1. Project Overview

HealthConnect Rwanda operates a network of 20 healthcare facilities delivering clinical, operational and financial services. Management requires clear visibility of patient volumes, clinical workloads, revenue collection, staff utilisation and patient experience. This report summarises the Power BI solution built for HealthConnect in Week 6 of the Experience Lab Internship. The solution has five report pages: Executive Overview, Patient Analytics, Clinical Analytics, Financial Analytics and Operational Analytics.

The report presents the main KPIs, what each page shows, answers to the core business questions in the brief, the items flagged during review, and ten strategic recommendations. Where a number was read from a chart instead of a card or table, it is marked with "~" and should be treated as approximate. SQL scripts, the data model and DAX measure documentation are covered in the accompanying project files.

2. Part A: Data and Dashboard Review

Scope. The model integrates appointments, billing, payments, insurance claims, laboratory tests, prescriptions, admissions, referrals, staff (doctors, nurses, radiographers, administrative staff), departments, equipment, beds and patient feedback. It covers 5,000 appointments linked to 4,443 unique patients across 20 facilities, 350 doctors and approximately 1,160 total staff. The analysis period spans January to December.

Consistency checks. Because the five pages draw on one model, their figures should agree. Key totals reconcile across pages as shown below.

Check	Result
Total Patients	4,443 on Executive Overview and Patient Analytics. Agree.
Total Appointments	5,000; visit-type split sums to 5,000 (≈1.65K Follow-up + ≈1.66K Consultation + ≈1.69K Emergency). Agree.
Total Revenue / Billed	771M RWF on Executive and Financial pages. Agree.
Cash Received	695M RWF; Collection Rate 90.07%; Outstanding 77M. Consistent.
Facilities / Doctors	20 facilities and 350 doctors on Executive and Operational pages. Agree.
Avg Satisfaction	2.93 / 5 on Executive Overview. Consistent with feedback volume.
Admissions	586 admissions; Rate 11.72%; Avg LOS 5.66 days. Internal consistency holds.
Lab / Prescriptions	2,272 lab tests and 4,003 prescriptions. Plausible relative to 5,000 appointments.

KPI definitions as built. Total Revenue is the sum of billed amounts across the 5,000 appointments. Cash Received is taken from payments.amount_paid. Collection Rate = Cash Received ÷ Total Billed. Admission Rate = Admissions ÷ Appointments. Average LOS is calculated only on admitted patients. Patient Satisfaction is the average of available feedback ratings (scale 1–5). Appointments per Doctor = Total Appointments ÷ 350 doctors.

3. Part B: KPI Summary

KPI	Value	Comment	Status
Total Patients	4,443	Unique patients linked to 5,000 appointments	Normal
Total Appointments	5,000	≈417/month; VisitType split nearly equal	Normal
Total Revenue (Billed)	771M RWF	Average bill ≈154,227 RWF	Normal
Cash Received	695M RWF	Collection Rate 90.07%; Outstanding 77M	Watch
Total Facilities	20	Even geographic spread across the network	Normal
Total Doctors	350	14.3 appointments per doctor on average	Normal
Total Staff	1,160	Nurses ≈700 (60%); Doctors 350 (30%)	Normal
Avg Patient Satisfaction	2.93 / 5	Below midpoint; feedback volume limited	Attention
Admission Rate	11.72%	586 admissions; Avg LOS 5.66 days	Normal
Abnormal Lab %	49.74%	Nearly half of lab results flagged abnormal	Watch
Claim Approval Rate	50.38%	Only half of claims approved; large pending stock	Attention
Appts per Doctor	14.3	Workload relatively even across departments	Normal
Appts per Bed	2.50	2,000 beds; utilisation appears moderate	Normal

Status is the author's judgement: **Attention** = management action needed now; **Watch** = needs definition or check; **Normal** = no concern from the dashboard.

4. Part C: Dashboard Findings

Executive Overview. The network handled 5,000 appointments for 4,443 patients and billed 771M RWF. Visit types are almost evenly split: Follow-up ≈33.9%, Consultation ≈33.1%, Emergency ≈33.0%. Monthly billed amounts fluctuate between ≈57M and ≈70M RWF, with a visible dip around September. Cardiology leads department revenue, followed by Radiology, Laboratory and Pediatrics. The weakest card is average patient satisfaction at 2.93/5.

Patient Analytics. One-time patients dominate (3,933); only 557 returning visits and 510 frequent visitors are recorded, giving an average of 1.13 visits per patient. Emergency appointments total 1,650. Diagnosis distribution is balanced across Hypertension, Gastritis, Upper Respiratory Infection, Diabetes and Malaria (each ≈19–21%). Severity is also balanced (Mild / Moderate / Severe each ≈33%). Facility patient volumes range from ≈200 (HF019) to ≈280 (HF013).

Clinical Analytics. 2,272 laboratory tests, 4,003 prescriptions and 586 admissions were recorded. Admission rate stands at 11.72% with average length of stay 5.66 days. Nearly half of lab results (49.74%) are abnormal. Top medications include Amlodipine, Omeprazole, Amoxicillin, Metformin and Artemether-Lumefantrine. Hypertension and Gastritis drive the largest appointment volumes; severity mix is even across the five main diagnoses.

Financial Analytics. Total billed 771M RWF; cash received 695M RWF (collection rate 90.07%); outstanding 77M RWF. Average bill is 154,227 RWF. Claim approval rate is only 50.38%. Payment methods are evenly split across Card, Cash, Insurance and Mobile Money (each ≈24–26%). Facility billed amounts range from ≈35M (HF018) to ≈44M (HF011). Cardiology remains the top revenue department. Insurance providers MMI, RSSB, Radiant and Private show similar claimed amounts, with roughly half of claims still pending.

Operational Analytics. 1,160 total staff (≈700 nurses, 350 doctors, remainder administrative and radiographers). 2,000 beds and 200 equipment items are recorded. Appointments per doctor average 14.3; appointments per bed average 2.50. Cardiology, Radiology and Obstetrics & Gynecology show the highest workload per doctor. Department appointment counts are tightly clustered (488–544), indicating balanced demand. Staff-by-role pie confirms nurses form the majority workforce.

5. Part D: Business Analysis

1. Which hospitals serve the highest number of patients? HF013, HF014, HF001, HF010 and HF011 lead on unique patient counts (approximately 250–280 each). The lowest volumes sit at HF019, HF005 and HF004 (≈200–220). Volume variation across the 20 facilities is moderate.

2. Which departments experience the greatest workload? Cardiology (544 appointments, 15.5 appts/doctor), Radiology (525, 15.0) and Obstetrics & Gynecology (511, 14.6) top the list. ICU is lowest on appointments per doctor (13.3). Workload is relatively even network-wide.

3. Which diseases are becoming more common? The five recorded diagnoses (Hypertension, Gastritis, Upper Respiratory Infection, Diabetes, Malaria) each account for roughly 19–21% of appointments. No single disease dominates; the mix is stable across months.

4. Are medical staff distributed efficiently? 350 doctors support 5,000 appointments (14.3 per doctor). Nurses form ≈60% of the 1,160 staff. Facility-level doctor counts sit between 16 and 19, and the scatter of doctors vs appointments shows no extreme outliers. Distribution appears broadly efficient.

5. Which insurance companies generate the largest revenue? MMI, RSSB, Radiant and Private each contribute similar claimed amounts (≈90–110M range). Approval rates are close to 50% across providers, so pending claims are a shared issue rather than provider-specific.

6. Which facilities have the highest patient satisfaction? Overall average is 2.93/5. Facility-level satisfaction rankings are not fully visible on the current pages; the network-wide figure is the primary signal and indicates room for improvement.

7. Which hospitals require expansion? Facilities with high patient volume and high admission counts (HF013, HF014, HF001) together with those showing elevated abnormal lab rates or longer LOS warrant capacity review. Bed utilisation (2.50 appts/bed) does not yet signal critical shortage.

8. How can operational costs be reduced? Raise the claim approval rate (currently 50.38%), accelerate collection of the 77M outstanding, reduce the share of abnormal lab results that may drive unnecessary repeat testing, and keep doctor workload balanced to avoid overtime or locum costs.

9. Which services contribute most to revenue? Cardiology leads, followed by Radiology, Laboratory, Pediatrics, Obstetrics & Gynecology, Internal Medicine, Surgery, Emergency, ICU and Pharmacy. The top four departments account for a substantial share of the 771M billed.

10. Which regions require additional healthcare investment? Geographic (province/district) patient and revenue maps are limited by the absence of a complete patient-to-village master in the supplied extract. Facility-level volume and revenue rankings are the best available proxy; higher-volume sites should be prioritised for equipment and staffing investment.

6. Part E: Flagged Items

The items below were found while reviewing the five pages. Priority reflects the effect on decisions: Critical and High items should be dealt with first.

Ref	Page	What was found	Action	Priority
F1	Financial	Claim approval rate is only 50.38%. Roughly half of all claims remain pending or unapproved.	Investigate rejection reasons; set a target approval rate and weekly claim ageing review.	Critical
F2	Executive	Average patient satisfaction is 2.93/5. Feedback volume appears limited relative to 5,000 appointments.	Increase feedback capture rate; add facility and department satisfaction rankings.	High
F3	Financial	77M RWF outstanding (≈10% of billed). Collection rate 90.07% is good but leaves material cash uncollected.	Age the outstanding balance; escalate bills older than 60/90 days.	High
F4	Clinical	49.74% of laboratory results are abnormal. This may indicate true disease burden or testing protocol issues.	Review lab ordering protocols and abnormal-result follow-up pathways.	High
F5	Patient	Only 557 returning visits and 510 frequent visitors out of 4,443 patients (avg 1.13 visits). Continuity of care appears low.	Introduce chronic-care pathways and recall systems for Hypertension, Diabetes and Malaria.	High
F6	All pages	No complete patient master (gender, age, blood group, province) was supplied. Several patient KPIs cannot be calculated.	Load the optional hc.patients table or source a patient demographic extract.	Medium
F7	Executive	Monthly billed amounts show a September dip. Confirm whether this is seasonal or a data-cut issue.	Confirm last data date; mark partial months if required.	Medium
F8	Operational	Staff-to-patient and bed-utilisation ratios are network averages only. Facility-level capacity stress is not visible.	Add facility-level bed occupancy and staff-to-patient ratio cards.	Medium
F9	Clinical	Admission rate 11.72% and Avg LOS 5.66 days are network-wide. Variation by diagnosis and facility is only partially shown.	Add diagnosis-level and facility-level LOS and admission rate visuals.	Medium
F10	Financial	Payment method split is even (Card/Cash/Insurance/Mobile Money). No trend or facility-level method preference is shown.	Add payment-method trend and facility breakdown for treasury planning.	Low
F11	Operational	Equipment items = 200; no availability or downtime measure is present.	Add equipment status (available / under maintenance) if the source supports it.	Low
F12	All pages	Not yet shown: gender/age/blood-group distributions, province maps, readmission rate, cancellation rate, and doctor-level satisfaction.	Prioritise these for the next dashboard iteration.	Medium

7. Part F: Strategic Recommendations

No.	Area	Based on	Recommendation
1	Claims management	50.38% approval rate; large pending stock (F1)	Agree a target approval rate with finance and clinical leads. Review rejected claims weekly by provider and reason. Clear the pending backlog within 60 days.
2	Patient experience	Avg satisfaction 2.93/5; limited feedback volume (F2)	Raise feedback capture to at least 40% of appointments. Publish facility and department league tables monthly and act on the bottom quartile.
3	Cash collection	77M outstanding; 90% collection rate (F3)	Age the receivables, prioritise bills older than 60 days, and introduce automated SMS reminders linked to the billing system.
4	Clinical quality	49.74% abnormal lab results (F4)	Audit high-volume tests (Malaria, CBC, Glucose, Urinalysis). Confirm whether ordering protocols or true disease burden drives the abnormal rate.
5	Continuity of care	Avg 1.13 visits per patient; low returning visits (F5)	Build chronic-disease registers for Hypertension, Diabetes and Malaria. Introduce structured follow-up appointments and community recall.
6	Data completeness	Missing patient master (gender, age, province) (F6)	Load the patient demographic extract into hc.patients so that gender, age-group, blood-group and geographic KPIs become available.
7	Capacity planning	Facility volumes 200–280 patients; 2.50 appts/bed (F8)	Model bed occupancy and staff-to-patient ratios at facility level. Flag sites approaching capacity for expansion or staffing support.
8	Department focus	Cardiology, Radiology, Lab and Pediatrics lead revenue and volume	Protect capacity in the top four revenue departments. Review equipment and specialist coverage for Cardiology and Radiology first.
9	Performance monitoring	KPI cards and cross-page consistency already strong	Name a dashboard owner, schedule a monthly KPI review with facility managers, and keep delivered/cash KPIs as the headline metrics.
10	Service growth	Balanced diagnosis mix; even visit-type split; strong collection	Use the stable demand base to pilot telemedicine follow-ups and pharmacy outreach for chronic patients, reducing pressure on physical visits while protecting revenue.

8. Summary

The HealthConnect dashboards bring appointments, clinical activity, revenue, claims, staff and patient experience into one coherent view. Key figures (4,443 patients, 5,000 appointments, 771M RWF billed, 695M cash received, 20 facilities, 350 doctors) reconcile across the five pages.

The network runs at a steady operational tempo: visit types are balanced, department workloads are even, and the collection rate of 90% is solid. The main pressure points are a claim approval rate of only 50%, an average satisfaction score of 2.93/5, 77M RWF still outstanding, and nearly half of laboratory results flagged abnormal. Continuity of care is low (1.13 visits per patient), which limits both clinical outcomes and lifetime revenue per patient.

Twelve items were flagged, five of them rated Critical or High. The immediate priorities are to clear the claims backlog, raise feedback capture and satisfaction, age and collect the outstanding balance, and complete the patient demographic layer so that gender, age and geographic KPIs can be reported. With those foundations in place, HealthConnect can move from descriptive dashboards to predictive capacity and risk models in subsequent phases.

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